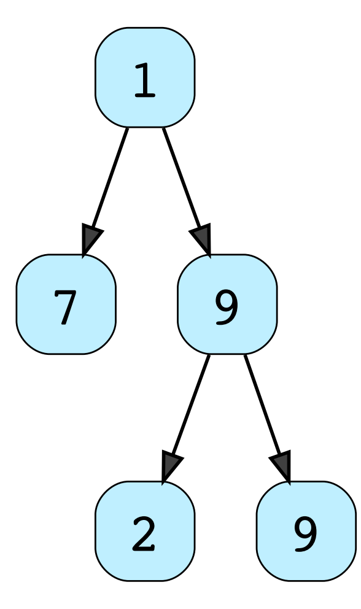
Solution: Sum of Path Numbers

Problem Statement

Given a binary tree where each node can only have a digit (0-9) value, each root-to-leaf path will represent a number. Find the total sum of all the numbers represented by all paths.

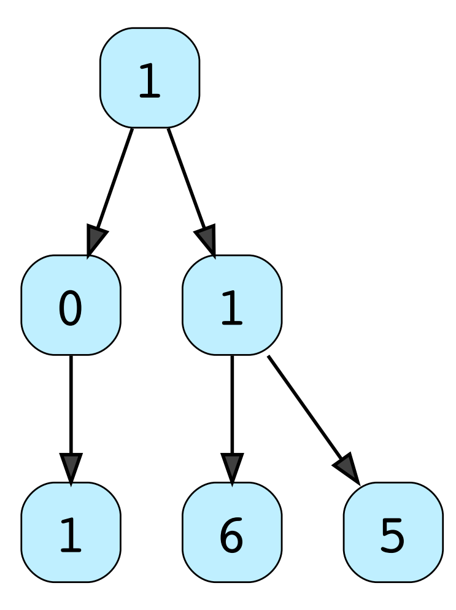
Example 1:

Output: 408
Explanation: The sum of all path numbers: 17 + 192 + 199



Example 2:

Output: 332
Explanation: The sum of all path numbers: 101 + 116 + 115

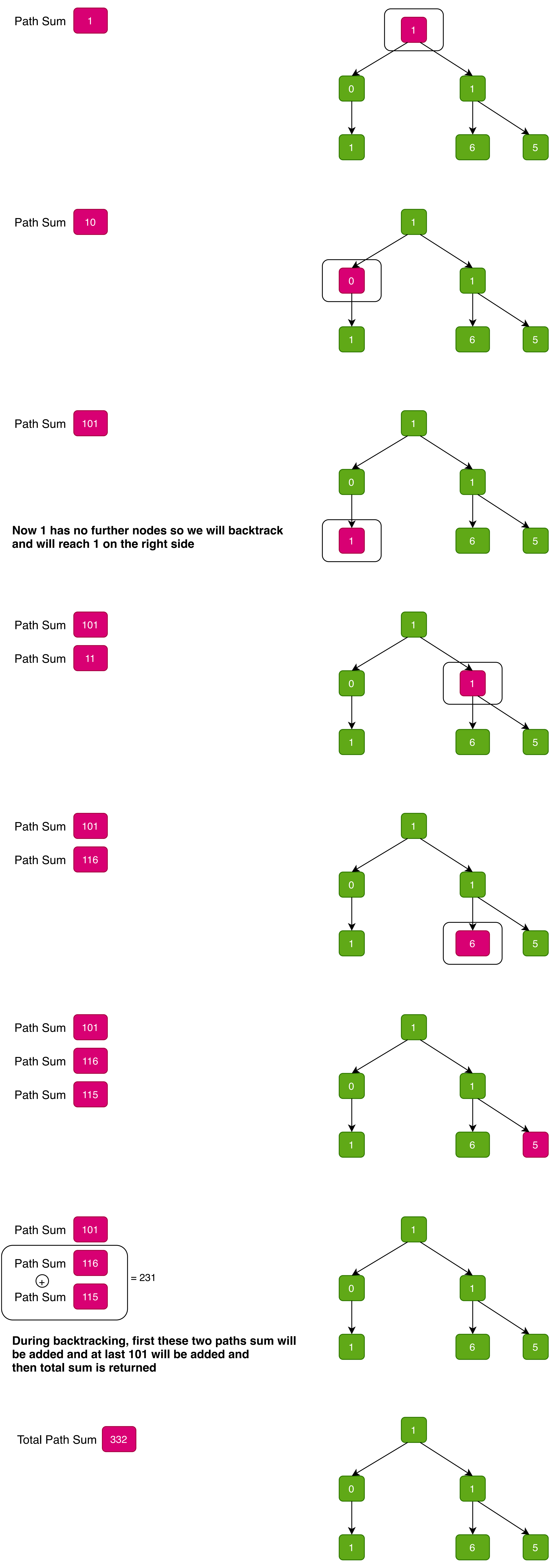


Solution

This problem follows the **Binary Tree Path Sum** pattern. We can follow the same **DFS** approach. The additional thing we need to do is to keep track of the number representing the current path.

How do we calculate the path number for a node? Taking the first example mentioned above, say we are at node ‘7’. As we know, the path number for this node is ‘17’, which was calculated by: `1 * 10 + 7 => 17`. We will follow the same approach to calculate the path number of each node.

Here is the visual representation of the algorithm:



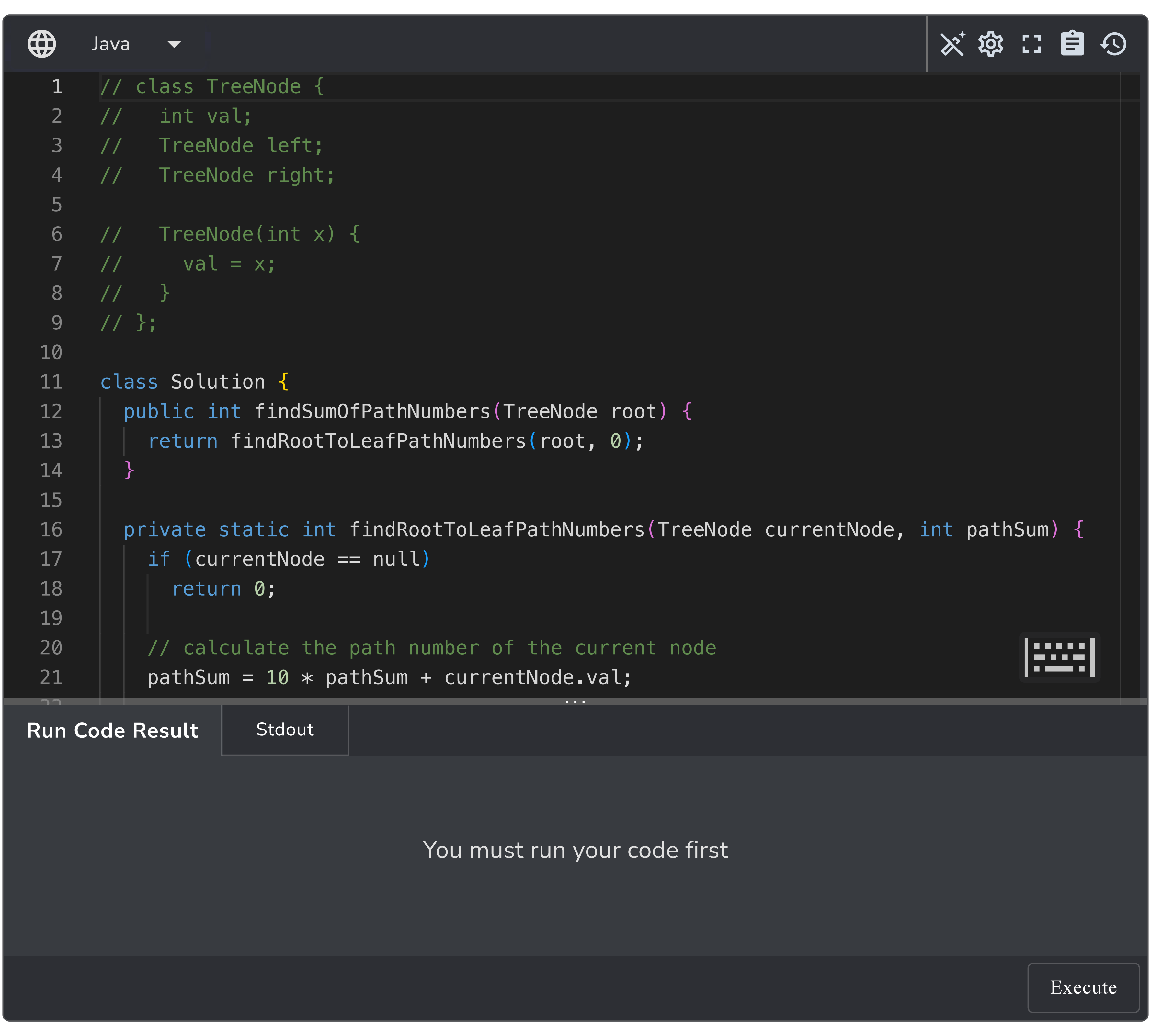
Now 1 has no further nodes so we will backtrack and will reach 1 on the right side

During backtracking, first these two paths sum will be added and at last 101 will be added and then total sum is returned

Sum of Path Nums

Code

Here is the code for this algorithm:



```java
// class TreeNode {
//   int val;
//   TreeNode left;
//   TreeNode right;

//   TreeNode(int x) {
//     val = x;
//   }
// };

class Solution {
  public int findSumOfPathNumbers(TreeNode root) {
    return findRootToLeafPathNumbers(root, 0);
  }

  private static int findRootToLeafPathNumbers(TreeNode currentNode, int pathSum) {
    if (currentNode == null)
      return 0;

    // calculate the path number of the current node
    pathSum = 10 * pathSum + currentNode.val;
```

You must run your code first

Time Complexity

The time complexity of the above algorithm is O(N), where ‘N’ is the total number of nodes in the tree. This is due to the fact that we traverse each node once.

Space Complexity

The space complexity of the above algorithm will be O(N) in the worst case. This space will be used to store the recursion stack. The worst case will happen when the given tree is a linked list (i.e., every node has only one child).

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Sum Of Path Numbers (Medium)

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Merge Intervals (Medium)

☐ Mark as Completed